Exponential & Logarithmic Functions: Compound Interest & LN

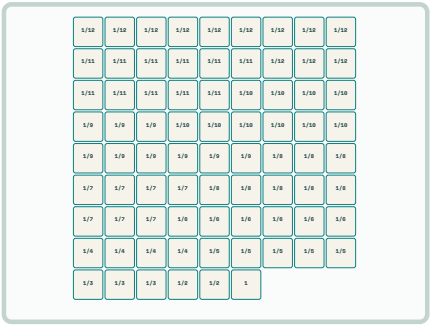
Tactile Concrete-Pictorial-Abstract (CPA) Station Guide • CCSS.MATH.HSF.LE.A.4

Student Name: _____ Date: _____ **Model: SC-32** **Check Score: ___ / 4**

STUDENT SELF-CHECK RULE (NO CHEATING):
Build with blocks and solve on paper first! Use your SC-32 calculator ONLY to verify answers on the 4-line LCD screen.

STATION LAB TUTORIAL: BENCH SCHEMATIC & CPA PROGRESSION

1. 2D Bench Blueprint



Kit: Fraction Blocks

2. Desk Setup (Concrete -> Pictorial)

Physical Bench Setup:
Represent exponential doubling by stacking proportional Fraction Studio blocks (7 mm, 14 mm, 28 mm, 56 mm).

Sketch Model on Paper:
Draw exponential growth curve showing rapid upward curvature.

3. Step-by-Step LCD Stack Trace (SC-32)

MODE:	123	Base-10 Arithmetic	
1.	1.05	X: 1.05	Growth factor 1.05 i
2.	ENTER	Y: 1.05 | X: 1.05	Stack lift
3.	10	Y: 1.05 | X: 10	Exponent t = 10 into
4.	y⊠	X: 1.6289	Power: 1.05^{10} = 1.62

EXPECTED LCD X: $1,628.89

1 **Calculate $500 at 6% for 5 years: 500 × $(1.06)^5$.**

Manipulative Model & Desk Sketch (Show your work):

Final Math Answer: _____

☐ Verified on SC-32

2 **Calculate $2,000 at 4% for 8 years: 2000 × $(1.04)^8$.**

Manipulative Model & Desk Sketch (Show your work):

Final Math Answer: _____

☐ Verified on SC-32

3 **Evaluate natural log ln(100) on SC-32: [100] [LN].**

Manipulative Model & Desk Sketch (Show your work):

Final Math Answer: _____

☐ Verified on SC-32

4 **Evaluate log base 10 of 10,000 on SC-32: [10000] [LOG].**

Manipulative Model & Desk Sketch (Show your work):

Final Math Answer: _____

☐ Verified on SC-32

Station Diagnostic & Solution Key: Chapter 6

Exponential & Logarithmic Functions: Compound Interest & LN • Manipulative Setup, Diagnostic Rubric & Verification Flow

MANIPULATIVE HARDWARE STATION INVENTORY & 3D PRINT PREPARATION

Station Kit: Fraction Studio Doubling Towers (FRAC-01) & SC-32 Stacks

CAD Blueprint: fraction_blocks

Target Device: SC-32 Scientific

Source File:

Education/scad/fraction_studio_parts.scad

Standard: Gridfinity 42mm Pitch / 0.6mm Inset Markings

Recommended Filaments:

Teal 3x1x6U Case & Proportional Fraction Towers (PLA)

#	Math Problem Prompt	Expected Math Answer	Self-Check Keystrokes & RPN Stack
1	**Calculate $500 at 6% for 5 years: 500 × (1.0** Standard: CCSS.MATH.HSF.LE.A.4	$669.11	[1][.][0][6] [ENTER] [5] [y^x] [5][0][0
2	**Calculate $2,000 at 4% for 8 years: 2000 × (** Standard: CCSS.MATH.HSF.LE.A.4	$2,737.14	[1][.][0][4] [ENTER] [8] [y^x] [2][0][0
3	**Evaluate natural log ln(100) on SC-32: [100]** Standard: CCSS.MATH.HSF.LE.A.4	4.6052	[1][0][0] [LN]
4	**Evaluate log base 10 of 10,000 on SC-32: [10** Standard: CCSS.MATH.HSF.LE.A.4	4.0000	[1][0][0][0][0] [LOG]

TEACHER DIAGNOSTIC GUIDE: COMMON MISCONCEPTIONS & REMEDIATION

Key Misconception to Target During Station Observation:

Multiply by principal P AFTER computing the exponential power (1+r)^t, not before.

How the Manipulatives & Calculator Foster Self-Correction:

Students solve with pencil and physical blocks first. When they self-check on the SC-32, any discrepancy between their written answer and the calculator's 4-line LCD immediately prompts them to re-inspect their physical bench model or regrouping, turning errors into productive struggle without teacher intervention.

Station Facilitation & Pacing Tip:

Keep 3D printed trays docked on standard Gridfinity baseplates to prevent loose parts on student desks. Encourage students to articulate their CPA steps: name the block representation (Concrete), point to their drawn tape/bond model (Pictorial), and verify register values (Abstract).

CLASSROOM SCORING & STATION MASTERY RUBRIC

Exemplary (18–20 pts):	Complete, accurate manipulative sketches; exact math answers; independently verified self-check on LCD stack.
Proficient (14–17 pts):	Correct math answers; minor sketch omissions; verified on calculator with minimal teacher prompting.
Developing (10–13 pts):	Understands problem concept but made computational slips; successfully self-corrected using calculator LCD.
Intervention (<10 pts):	Relied on calculator before solving on paper, or struggled with underlying physical manipulative models.